



Alcohol speeds up your heart

A new study indicates that drinking alcohol will increase your heart rate. The more you drink, the higher it will be. <http://digit.in/HicHic>



Omega-6 =| Premature death

Researchers have found that a high level of Omega-6 acids will reduce your chance of premature death. <http://digit.in/OM6Elixir>

An eco-friendly satellite propulsion system

Using water as a propellant for satellite propulsion might soon become a reality.

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Travelling to outer space had been our wet dream and we finally landed on the moon. The dream was fulfilled, but it has now been extended to

Mars. However, we are far from it and might not see that happening even in the next decade. Chemical rocket systems have a huge fuel requirement and takes up majority of the weight of the rocket. This propulsion system is mostly used for rocket launch vehicles but we'll be focussing on satellite propulsion systems in this story.

Satellites don't require high thrust. Hence, electrothermal and ion thrusters become the standing candidates. Research is still going on to improve the efficiency of these existing systems. A space company in India thinks otherwise. Bellatrix Aerospace, a private company in India, is experimenting with microwave electrothermal thrusters (MET) that uses water as its fuel.

WHAT IS BELLATRIX AEROSPACE?

Bellatrix Aerospace primarily deals with satellite propulsion systems including microwave electrothermal thrusters, Hall Effect thrusters and nano thrusters. Their main focus is their Microwave Electrothermal Thruster (patent pending) for satellites that will run

on water as a propellant. Back in 2011, Rohan Ganapathy, an aeronautical engineer from Hindustan College of Engineering and Technology, Coimbatore, was busy researching the MET thruster technology. With some progress, Yashas Karanam, an electrical engineer from Sri Jayachamarajendra College of Engineering, Mysore, also joined him. Both of them had a vision of turning the forgotten propulsion system into reality. After receiving a grant of Rs. 20 lakhs from JSW in 2013, they finally came out with a proof of concept by the end of the year 2015.

The company was incubated at the Indian Institute of Science (IISc) in Bengaluru. All the lab facilities and faculty support has been provided by the institute. Apart from working on satellite propulsion systems, the other vertical of Bellatrix includes launch vehicles.

For now, they are focussing on the microwave electrothermal thruster project for which they have already received an order from ISRO for their future satellite missions. Bellatrix Aerospace claims to be the only private company in India working on satellite propulsion systems. They are funded by angel investment and they are actively seeking to raise funding on a much bigger scale for their ambitious future projects.

EFFICIENT PROPULSION SYSTEM

Before considering the microwave electrothermal thruster, the duo was working on magnetoplasmadynamic thrusters. However, that was short-lived because this particular system required a huge amount of power, which can't be generated on-board a satellite.

Let's briefly understand how the MET thruster works. The basic idea here is to inject microwave energy into a cavity containing plasma where the microwave energy will be absorbed and dissipated in the form of thermal energy. An electromagnetic resonant cavity creates an alternating electromagnetic field where free-floating plasma is created and sustained. Along with the microwave energy and propellant gas, all the free electrons and heavy particles start colliding with each other. This increases the temperature of the propellant and accelerates it through the nozzle to generate thrust. In Bellatrix's design, they were able to build this system and successfully run it using water vapour as propellant. Water is easy to store, transport, and easily available. Hence, the application of this thruster in satellites can be a huge deal. A patent for this, applied in late 2016, is yet to be granted.



Rohan Ganapathy,
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